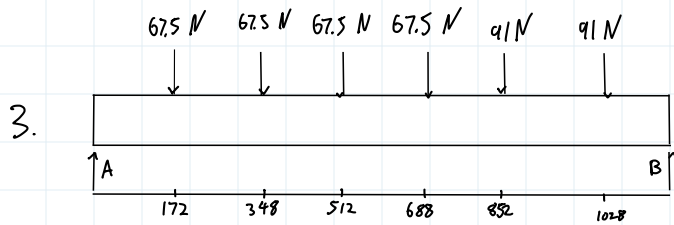




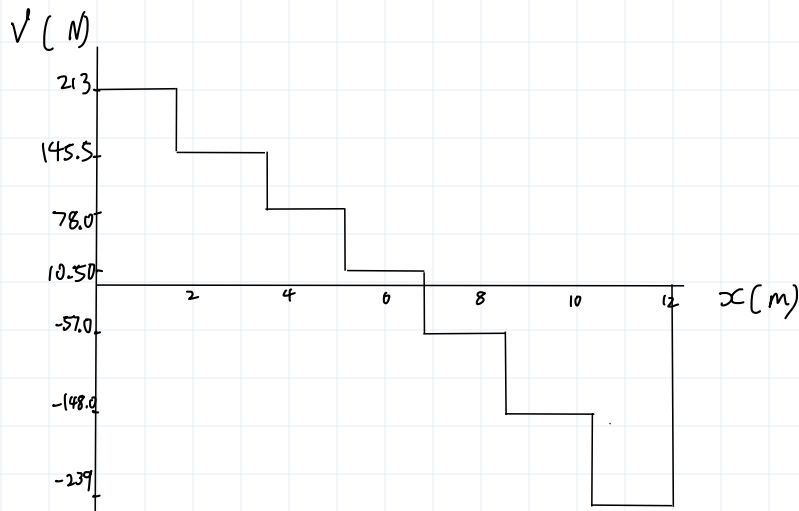
CIV102 Project Deliverable #1
Team 107 TA: Christian



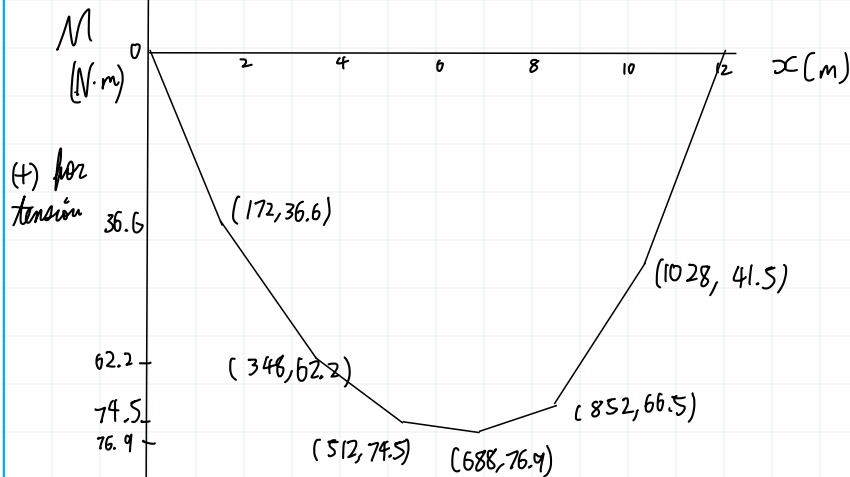
$$\sum M_A = 0 \Rightarrow B = 239 \text{ N}$$

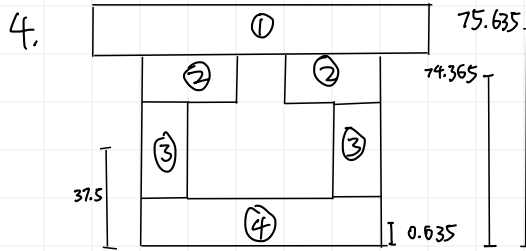
$$\sum F_y = 0 \Rightarrow A = 213 \text{ N}$$

SFD:



BMD:





i	$\bar{y}_i$ (mm)	A_i (mm ²)	I_i (mm ⁴)
1	75.635	127	204.8383
2	74.365	15.9258	25.686723
3	37.5	18.0484	966337.22
4	0.635	101.6	163.87064

$$\bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i} = \boxed{41.431}$$

$$\begin{aligned}
 I &= \sum (I_i + A_i (\bar{y}_i - \bar{y})^2) \\
 &= 80561 + \sum A_i (\bar{y}_i - \bar{y})^2 \\
 &= 418352 \text{ mm}^4 \\
 &= \boxed{4.18 \times 10^5 \text{ mm}^4}
 \end{aligned}$$

5. $\sigma = \frac{My}{I}$

$$M_{\max} = 76.9 \text{ N}\cdot\text{m} = 76.9 \times 10^3 \text{ N}\cdot\text{mm}$$

tension: bottom

$$\begin{aligned}
 y_{\max} &= 41.4 \text{ mm} \\
 \sigma_c' &= \frac{M y}{I} = \frac{(76.9 \times 10^3 \text{ N}\cdot\text{mm})(41.4 \text{ mm})}{(4.18 \times 10^5 \text{ mm}^4)} = \boxed{3.94}
 \end{aligned}$$

compression: top

$$\begin{aligned}
 y_{\max} &= 75 + 1.27 - \bar{y} = 34.8 \text{ mm} \\
 \sigma_c &= \frac{M y}{I} = \frac{(76.9 \times 10^3 \text{ N}\cdot\text{mm})(34.8 \text{ mm})}{(4.18 \times 10^5 \text{ mm}^4)} = \boxed{0.937} \rightarrow \text{will fail!}
 \end{aligned}$$